

TrES-1b

R-0173

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_181019 · created 2026-07-16T18:01:14+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458410.6469 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.01
Transit depth (Rp/Rs)^2	3.06 +/- 0.36 [%]
Orbital Inclination (inc)	89.96 +/- 0.63
Airmass coefficient 1 (a1)	0.911 +/- 0.014
Airmass coefficient 2 (a2)	0.0573 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#2 - [584, 311]
Optimal Aperture	3.04
Optimal Annulus	7.3
Transit Duration (day)	0.1074 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.175 $\pm$ 0.01 vs 0.1358 (deviation 3.02 σ)
Duration fit vs published	0.1074 d vs 0.1042 d (deviation 2.07 σ)
Mid-time O–C	-1.1 min
Depth SNR	13.5
Residual scatter	1.6 %

Light curve

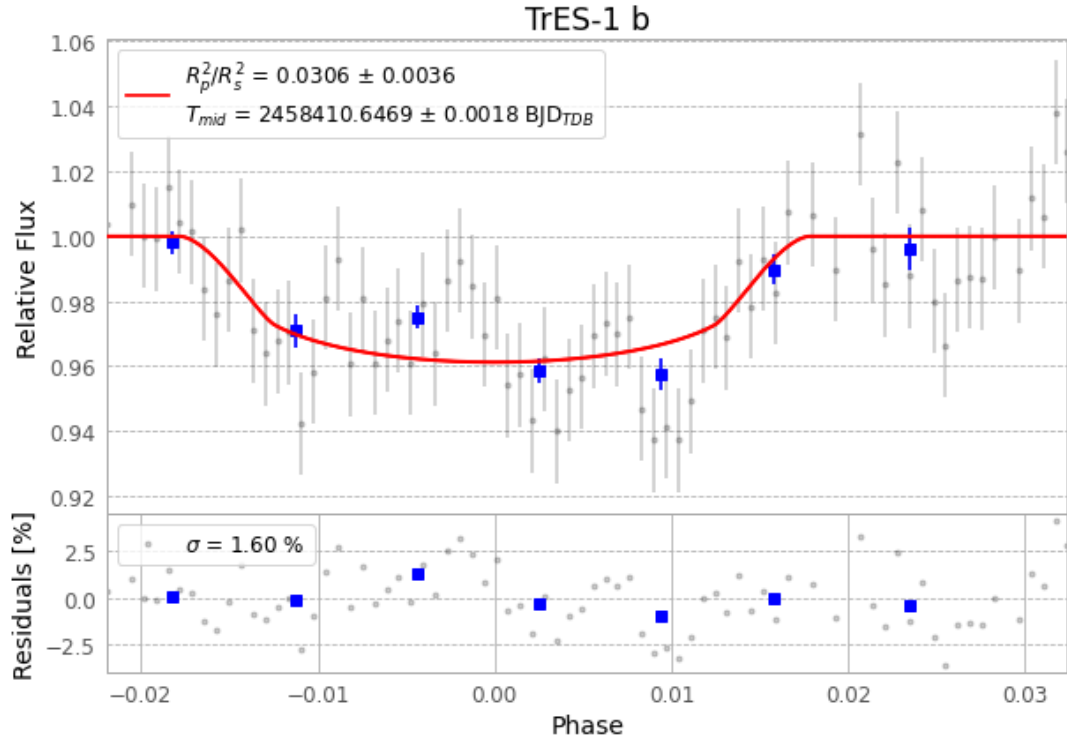


Figure 1 — FinalLightCurve_TrES-1 b_2018-10-18.png (SHA-256 8c73042030c6ac3a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 335], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8b27c00169413823...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9b8a76a

Data provenance

79 FITS frames (52 MB), TRES-1181019015412.FITS → TRES-1181019055112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-181019.log (d32fc08905ce...), FinalLightCurve_TrES-1 b_2018-10-18.png (8c73042030c6...), FinalLightCurve_TrES-1 b_2018-10-18.csv (478a04ff1976...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 18:01Z.